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Roll No

CS-222-CBCS
B.E., III Semester
Examination, June 2020
Choice Based Credit System (CBCS)
Digital Circuit and Design
Time : Three Hours

Maximum Marks : 60

Note: i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Sketch the diagram neatly, if any.

1. a) Explain multiple output switching function.
b) State and prove basic law of Algebra.
2. a) What is Universal gate? Implement AND, OR, NOT gate using NAND gates.
b) Design and draw full adder circuits.
3. a) Explain working of PMOS, NMOS and CMOS logic.
b) Explain astable multivibrator.
4. a) Explain counters. Also differentiate between synchronous and asynchronous counter.
b) Write short note on shift registers.
5. a) What is the need of analog to digital converter? In detail explain any one digital converter.
b) Write short note on Sample and hold circuits.

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6. a) Simplify the Boolean function using K-map
$$F(A, B, C, D) = \sum(0, 1, 2, 4, 5, 6, 8, 9, 10, 12, 13, 14)$$

b) Explain BCD adder.
7. a) Explain Schmitt trigger circuit.
b) Explain semiconductor memories.
8. Write short notes on-
 - i) De-Morgan's law
 - ii) Latch (NAND and NOR)
 - iii) Astable multivibrator
 - iv) Ring counter
